Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 1 of 7 · Question 6 part c · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

6(a)	short term is recent event/long term goes back many years; short term very little processing, long term has more processing;		**(2)**
Question Number	Answer	Additional guidance	Mark
6(b)	cerebral hemispheres/cerebral cortex/frontal lobe/cerebrum;		**(1)**
Question Number	Answer	Additional guidance	Mark
6(c)	• language/speaking/talking problems / repetitive actions/confusion; • loss of voluntary muscle function; • loss of intellectual capacity/function; • deterioration in social behaviour;		**(2 max)**
Question Number	Answer	Additional guidance	Mark
6(d)	• blocks synapse/cleft/build-up of plaques; • prevents diffusion of acetylcholine / transmitter/neurones communicating; • blocks post-synaptic receptors/prevents generation of new impulse; • death of neurones;		**(3 max)**

Question Number	Answer	Additional guidance	Mark
6(e)	• inhibits/prevents breakdown of acetylcholine / transmitter; • increases chance of synaptic transmission / neurones communicating/generating new impulse; • reduces death of neurones;		**(2)**
Question Number	Answer	Additional guidance	Mark
6(f)	formation of β-amyloid is caused by incorrect breakdown of APP; build-up/formation of β-amyloid (in neurones);		**(2)**

Total 12 Marks

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 1 of 7 · Question 6 part c · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 2 of 7 · Question 6 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

6(a)	short term is recent event/long term goes back many years; short term very little processing, long term has more processing;		**(2)**
Question Number	Answer	Additional guidance	Mark
6(b)	cerebral hemispheres/cerebral cortex/frontal lobe/cerebrum;		**(1)**
Question Number	Answer	Additional guidance	Mark
6(c)	• language/speaking/talking problems / repetitive actions/confusion; • loss of voluntary muscle function; • loss of intellectual capacity/function; • deterioration in social behaviour;		**(2 max)**
Question Number	Answer	Additional guidance	Mark
6(d)	• blocks synapse/cleft/build-up of plaques; • prevents diffusion of acetylcholine / transmitter/neurones communicating; • blocks post-synaptic receptors/prevents generation of new impulse; • death of neurones;		**(3 max)**

Question Number	Answer	Additional guidance	Mark
6(e)	• inhibits/prevents breakdown of acetylcholine / transmitter; • increases chance of synaptic transmission / neurones communicating/generating new impulse; • reduces death of neurones;		**(2)**
Question Number	Answer	Additional guidance	Mark
6(f)	formation of β-amyloid is caused by incorrect breakdown of APP; build-up/formation of β-amyloid (in neurones);		**(2)**

Total 12 Marks

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 2 of 7 · Question 6 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 3 of 7 · Question 6 part f · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part f in this mark scheme extract

6(a)	short term is recent event/long term goes back many years; short term very little processing, long term has more processing;		**(2)**
Question Number	Answer	Additional guidance	Mark
6(b)	cerebral hemispheres/cerebral cortex/frontal lobe/cerebrum;		**(1)**
Question Number	Answer	Additional guidance	Mark
6(c)	• language/speaking/talking problems / repetitive actions/confusion; • loss of voluntary muscle function; • loss of intellectual capacity/function; • deterioration in social behaviour;		**(2 max)**
Question Number	Answer	Additional guidance	Mark
6(d)	• blocks synapse/cleft/build-up of plaques; • prevents diffusion of acetylcholine / transmitter/neurones communicating; • blocks post-synaptic receptors/prevents generation of new impulse; • death of neurones;		**(3 max)**

Question Number	Answer	Additional guidance	Mark
6(e)	• inhibits/prevents breakdown of acetylcholine / transmitter; • increases chance of synaptic transmission / neurones communicating/generating new impulse; • reduces death of neurones;		**(2)**
Question Number	Answer	Additional guidance	Mark
6(f)	formation of β-amyloid is caused by incorrect breakdown of APP; build-up/formation of β-amyloid (in neurones);		**(2)**

Total 12 Marks

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 3 of 7 · Question 6 part f · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part f in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 4 of 7 · Question 5 part b.i · 2 marks

2023 JUN · Paper 02 (regional R) · find part b.i in this mark scheme extract

Number				
5	(a) (i)	Any two from: • chemical; • diffused from one neurone to another; • allows neurones to communicate / reference to communication between presynaptic and post synaptic neurones;		2
	(ii)	• neurotransmitter/dopamine/ diffuses/travels across synapse; • binds with receptor (on postsynaptic membrane)		2
	(iii)	Any two from: • increased movement; • increased heart rate/blood pressure; • more energy; • sleeping difficulties; • anxiety/stress; • hallucinations;		2
	(b) (i)	Any two from • tremors/random muscular movement / involuntary movement; • slower movement; • speech/writing changes; • impaired posture/balance; • sleeping problems; • rigidity/stiffness; • anxiety/depression/mental health/memory problems;		2
	(ii)	• (145 000 ÷ 100) (x 18) = 26 100; • (26 100 + 145 000) = 171 100;	Allow ecf for mp2 - if mp1 has an incorrect value but this value is added to 145 000 to arrive at the correct answer for the calculation award 1 mark Full marks for correct final answer	2

Total for question 5 = 10 marks

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 5 of 7 · Question 4 part c.iv · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part c.iv in this mark scheme extract

4 (a)	• diarrhoea by clean water/safe water/ bottled water/boiled water/ sewage treatment/chlorination/good sanitation/not mixing sewage with drinking water/washing hands/good hygiene/ cooking food thoroughly; • malaria by nets/draining swamps/insecticides/ insect repellent's/wearing clothing to cover arms (and legs/mosquito coils/anti-malarial/vaccination ;	Ignore ORT	2
(b) (i)	antibiotics;		1
(ii)	• elderly have weakened/lowered immune system/immunocompromised; • indoor spaces makes transmission easier/reduced air flow/ reduced ventilation/build- up of air droplets with pathogens/ people closer together;		2
(c) (i)	• diet; • exercise; • smoking: • drinking alcohol: • stress	Accept drugs	2
(ii)	• $\frac{5\,000\,000}{100\,000} = 50$; • 50 × 445 = 22 250; • 22 300; ***22,250 scores 2 marks*** ***22,300 scores 3 marks***	***1 mark for ×445 if there are no other marks***	3
(iii)	• use of 445 and 185; • $\frac{445}{185}$; • 2.41:1/2.405:!/2.4:1= 3 marks;	ecf from mp1 **if they have done the correct calculation and 2:1 is on the answer line they score** 2 marks	3
(iv)	• people live longer/elderly population; • these diseases are more common in older people; • less likely to die younger from infectious diseases/better medical facilities/treatment; • different diet/ more meat/saturated fats/processed food; • drinking more alcohol;	People live longer so there is more cancer and dementia/these diseases= 2 marks	2

Total 15 marks

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 5 of 7 · Question 4 part c.iv · 2 marks

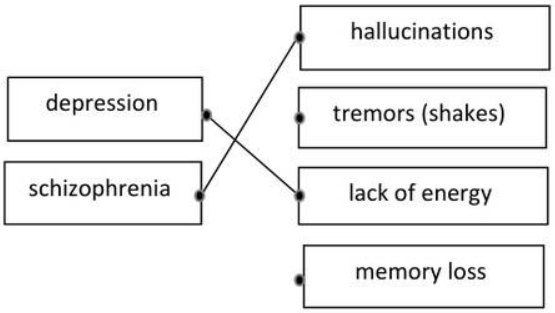
2024 NOV · Paper 02 · extract 2 of 2 · find part c.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 6 of 7 · Question 2 part a.iv · 1 mark

2022 JUN · Paper 02 · extract 1 of 2 · find part a.iv in this mark scheme extract

2 (a) (i)	2002;		1
(ii)	2005 (1.4)		1
(iii)	2.75:2.6; 1.1/1.06 :1;		2
(iv)	low income/low self esteem/(domestic) violence/low social status/substance abuse;		1
(b)			2
(c)	**A** alcohol and heroin; B antibiotics have no effect on mental health C paracetamol has no effect on mental health D antibiotics have no effect on mental health		1
(d)	Any two from the following brain/skin/sweat glands/muscles/blood vessels/arterioles;;		2
			Total 10

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 6 of 7 · Question 2 part a.iv · 1 mark

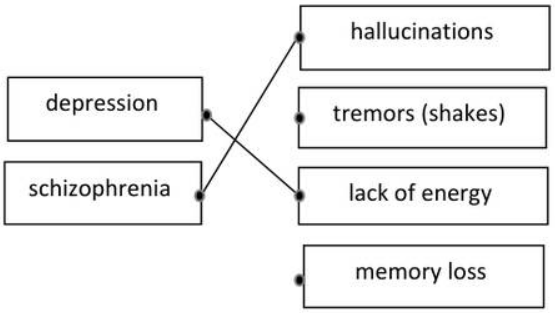
2022 JUN · Paper 02 · extract 2 of 2 · find part a.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 7 of 7 · Question 2 part b · 2 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

2 (a) (i)	2002;		1
(ii)	2005 (1.4)		1
(iii)	2.75:2.6; 1.1/1.06 :1;		2
(iv)	low income/low self esteem/(domestic) violence/low social status/substance abuse;		1
(b)			2
(c)	**A** alcohol and heroin; B antibiotics have no effect on mental health C paracetamol has no effect on mental health D antibiotics have no effect on mental health		1
(d)	Any two from the following brain/skin/sweat glands/muscles/blood vessels/arterioles;;		2
			Total 10

Type MS · State causes and symptoms of dementia, Parkinson's and mental illness

Answer 7 of 7 · Question 2 part b · 2 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks